0.1 eV ▼50 keV 10 MeV

# How does a tube voltage become a shortest wavelength?

? One electron, one photon, and a sharp edge that measures h/e.

## 1 Making X-rays is thermionic emission plus a cliff

A filament boils electrons off; tens of kilovolts accelerate them into a heavy target; they brake in the nuclear field and radiate. Only about one per cent of the electron power becomes X-rays — the rest heats the anode, which is why the anode rotates.

$$\eta \approx 10^{-9} ZV: \quad \text{W, } 100 \text{ kV} \Rightarrow 0.7\ \%$$

→ a 700 W dental tube radiates a few watts and must dump the other 695 W

## 2 The cutoff is an energy statement

A braking electron may shed any fraction of its energy, so the spectrum is continuous. But no photon can carry more than the whole electron energy, and that bound is reached when one electron gives everything in one collision.

$$hf_{\max} = eV \Rightarrow \lambda_{\min} = \frac{hc}{eV} = \frac{1.24 \times 10^6}{V\,[\text{V}]}\ \text{pm}$$

→ 30 kV → 41.3 pm; 50 kV → 24.8 pm; 100 kV → 12.4 pm

## 3 Why the edge is sharp and the tail is soft

The short-wavelength edge is a kinematic extreme: nothing can be shorter, so the distribution starts abruptly there. The long-wavelength side fades because soft photons are reabsorbed in the target and the window before they escape.

→ changing the target changes brightness roughly as Z, never the edge

## 4 A worked cutoff check

A tube runs at 50 kV. What is the shortest wavelength, and what does 100 kV do to it?

$$\lambda_{\min} = \frac{1.24 \times 10^6}{5 \times 10^4} = 24.8\ \text{pm}$$

$$100\ \text{kV}: \ 12.4\ \text{pm}$$

→ plot λ_min against 1/V and the slope is hc/e: the Duane-Hunt measurement of h

KEY RESULT

the tube in three lines

$$\lambda_{\min} = hc/eV$$

$$\eta \approx 10^{-9} ZV$$

edge from V, height from Z and mA

TRAP

two confusions

Applying λ_min to a characteristic line — it bounds only the continuous spectrum.

Thinking higher Z shifts the edge. It raises the hump.

NUMBERS

tube bench

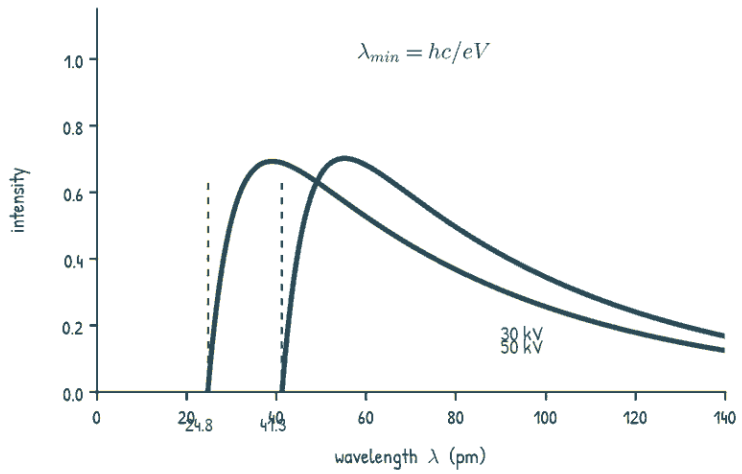
1.24×10⁶ pm·V rule

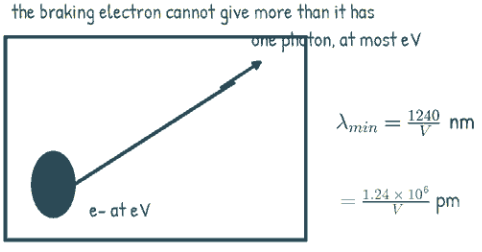
tungsten Z = 74, melting point 3422 °C

1 % efficiency at 100 kV

Duane-Hunt: the shortest X-ray is one electron spending everything at once

voltage sets the edge, target sets the height




target Z changes brightness, never the cutoff

KEEP / λ_min = hc/eV: the edge depends on voltage alone, never on the target




# When do the sharp lines appear?

? A spectrum has two parts with two different masters: voltage and the atom.

### 1 Two mechanisms in one spectrum

Braking radiation gives the continuous hump, cut off at hc/eV. Separately a fast electron can eject an inner one; the vacancy refills from above and the photon carries a level difference — a sharp, element-specific line.

$$E_{K\alpha} = E_L - E_K, \qquad E_{K\beta} = E_M - E_K$$

→ the lines sit where the atom says; the hump sits where the voltage says

### 2 The threshold logic examiners test

A K line needs a K vacancy, and making one costs at least the K binding energy. Below that voltage the lines are absent however large the current; above it they appear and grow.

$$V_{\text{th}} = \frac{E_K}{e}: \quad \text{Cu } 9\text{ kV},\ \text{Mo } 20\text{ kV},\ \text{W } 70\text{ kV}$$

→ no lines means below threshold, not a failed tube

### 3 Screened hydrogen, and Moseley

The falling L electron sees a nucleus of charge Z screened by the one remaining K electron, so the transition is hydrogenic arithmetic with (Z − 1) for Z. The square root linearises it: a straight line in Z — how Moseley ordered the elements.

$$E_{K\alpha} = \tfrac{3}{4}(13.6)(Z-1)^2 \text{ eV}, \qquad \sqrt{f} = \sqrt{\tfrac{3}{4}cR}\,(Z-1)$$

→ copper: 8.0 keV predicted, 155 pm against a measured 154 pm

### 4 A worked threshold

Copper's K shell binds 8.98 keV. What is the minimum tube voltage for its Kα line?

$$V_{\text{th}} = \frac{8.98\text{ keV}}{e} = 8.98\text{ kV}$$

$$V = 4.5\text{ kV}: \ \text{cutoff } \frac{1.24\times10^6}{4500} = 276\text{ pm},\ \text{no lines}$$

→ below threshold the continuous spectrum survives and the lines vanish: two masters, two behaviours

KEY RESULT
line and threshold

$$E_{K\alpha} = \tfrac{3}{4}(13.6)(Z-1)^2$$

$$\sqrt{f} \propto (Z-1)$$

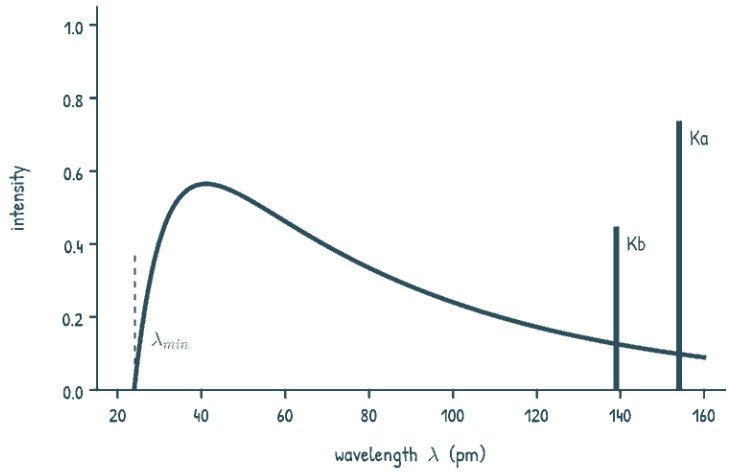
V_th = E_K/e; below it, no lines

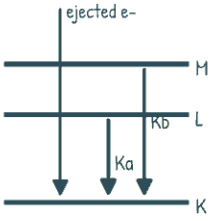
GATE
what the shielded form assumes
One K electron left, so it works for K lines of heavy atoms.
L lines need b ≈ 7.4, so the constant is line-specific.

TRAP
the two masters
Raising current brightens both parts; raising voltage moves the edge and can switch lines on. Neither moves a line's wavelength.

Characteristic lines are the atom fingerprint; Moseley turned them into a periodic table ruler

lines ride on the bremsstrahlung hump




$E_{K\alpha} = \tfrac{3}{4}(13.6)(Z-1)^2$ eV

$\sqrt{f} \propto (Z-1)$

Cu: K binds 9 keV, Kα = 8.0 keV

the vacancy must be affordable: V > 9 kV

KEEP / the hump is bremsstrahlung; the spikes need a vacancy, so they need a threshold voltage

0.1 eV — 100 keV — 10 MeV

# Why does a radiograph show bone and not skin?

? Attenuation is exponential, and the exponent is roughly Z cubed.

1

## A fixed fraction per layer

Every additional slice of material removes the same fraction of what is left, because it presents the same number of absorbers. That sentence is the differential equation, and its solution is the exponential.

$$I = I_0 e^{-\mu x}, \qquad x_{1/2} = \frac{\ln 2}{\mu}$$

→ lead at 100 keV: $x_{½}$ = 0.12 mm, so μ = 58 cm⁻¹

2

## Why bone casts a shadow

Photoelectric absorption scales roughly as $Z^3$–$Z^4$ and falls with photon energy. Calcium (Z = 20) against carbon and oxygen (Z = 6, 8) is therefore a contrast of order twenty in μ, which the film records as a shadow. Barium and iodine meals exploit the same appetite.

→ soft X-rays give better contrast and more dose: one curve read two ways

3

## The engineering trade

The same μ that creates contrast also deposits energy, and the energy deposited per unit mass is the dose. Thicker patients need harder beams, harder beams give less contrast, and the compromise is what a radiology protocol is. Computed tomography is this law inverted numerically over many projections.

$$\text{dose} \propto \int \mu \, \Phi \, dt$$

4

## A worked half-value stack

Lead at 100 keV has $x_{½}$ = 0.123 mm. What thickness leaves one per cent?

$$n = \frac{\log(0.01)}{\log(0.5)} = 6.64 \text{ half layers}$$

$$x = 6.64(0.123) = 0.82 \text{ mm}$$

→ a lead apron is 0.25–0.5 mm: enough for scattered diagnostic radiation, not for a beam

KEY RESULT

the law

$$I = I_0 e^{-\mu x}$$

$$x_{1/2} = \ln 2 / \mu$$

μ rises as $Z^3$–$Z^4$, falls with E

GATE

narrow beam only

μ is measured with a narrow beam and a narrow detector. A broad beam collects scattered photons and needs a build-up factor.

NUMBERS

half-value layers

Pb 0.12 mm at 100 keV

Al a few mm at 50 keV

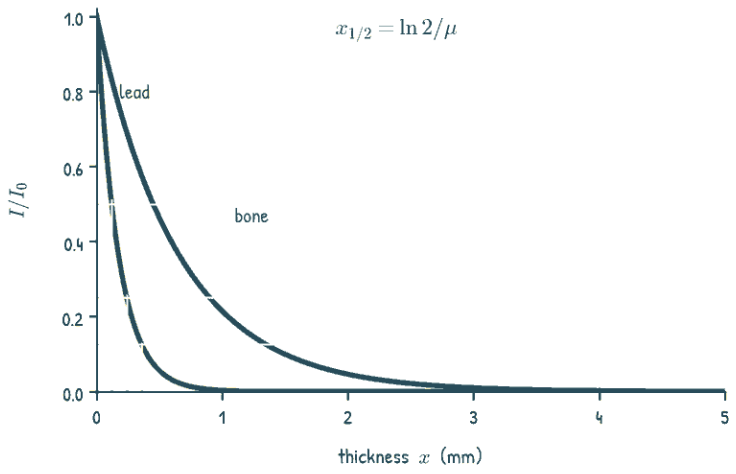
tissue ~ cm at 50 keV

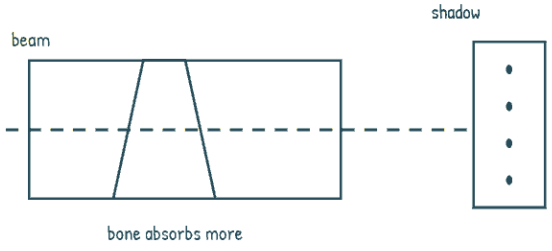
Exponential attenuation: the radiograph is a map of mu times thickness

every layer removes the same fraction

$x_{1/2} = \ln 2 / \mu$

$\mu \propto Z^3 - Z^4$, falls with $E$




bone absorbs more

contrast and dose are the same curve, read two ways

KEEP / I = I₀ e^{−μx}: contrast comes from μ, and μ is a steep function of Z and E

0.1 eV ▼8 keV 10 MeV

# How do you measure an atom with a wavelength?

? Bragg's law turns angles into spacings — and spacings into Avogadro's number.

### 1 Only a crystal is fine enough

A grating resolves structure of order its own spacing. Visible light (500 nm) sees atomic planes (0.3 nm) as a smooth surface — the ratio is $10^3$ too coarse for any Bragg angle. X-rays at 100 pm match the spacing, so the crystal becomes the grating.

→ Laue's 1912 spots proved both that X-rays are waves and that crystals are lattices

### 2 The path difference, with the angle made foolproof

Two rays strike successive planes at glancing angle $\theta$. The lower one travels an extra $d\sin\theta$ going in and the same coming out. Constructive interference needs that total to be a whole number of wavelengths.

$$2d\sin\theta = n\lambda, \qquad n \le \frac{2d}{\lambda}$$

→ $\theta$ is measured from the plane; the detector sits at $2\theta$ from the beam

### 3 From a spacing to Avogadro

In rock salt the cubic cell alternates ions, so each ion pair owns a volume $2d^3$ where d is the neighbouring-ion spacing. A balance gives the same volume as $M/\rho N_A$, and equating them yields Avogadro's number from a wavelength and an angle.

$$N_A = \frac{M}{2\rho d^3} = \frac{0.05844}{2(2165)(2.82\times10^{-10})^3}$$

$$= 6.02\times10^{23}\ \text{mol}^{-1}$$

→ the missing reflections are structure-factor information, not experimental error

### 4 A worked Avogadro

NaCl: d = 0.282 nm, ρ = 2165 kg/m³, M = 58.44 g/mol.

$$N_A = \frac{0.05844}{2(2165)(2.82\times10^{-10})^3} = \frac{0.05844}{9.71\times10^{-26}}$$

$$= 6.02\times10^{23}\ \text{mol}^{-1}$$

→ a wavelength, an angle and a balance reproduce a constant measured in electrochemistry

KEY RESULT

the law and its use

$$2d\sin\theta = n\lambda$$

$$d = \frac{n\lambda}{2\sin\theta}$$

$$N_A = M/2\rho d^3$$

TRAP

the convention

θ from the plane, so the path difference is 2d sinθ. Writing 2d cosθ is the single most common error in this topic.

And the detector angle is 2θ.

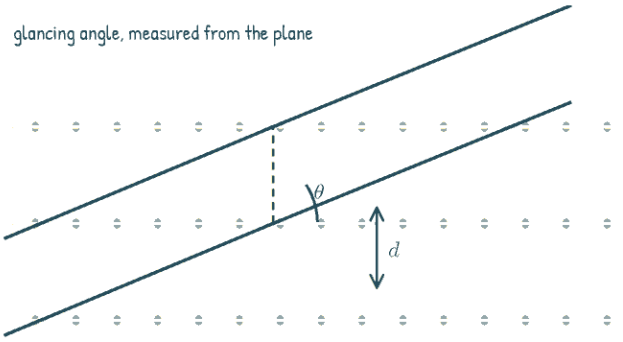
NUMBERS

NaCl bench

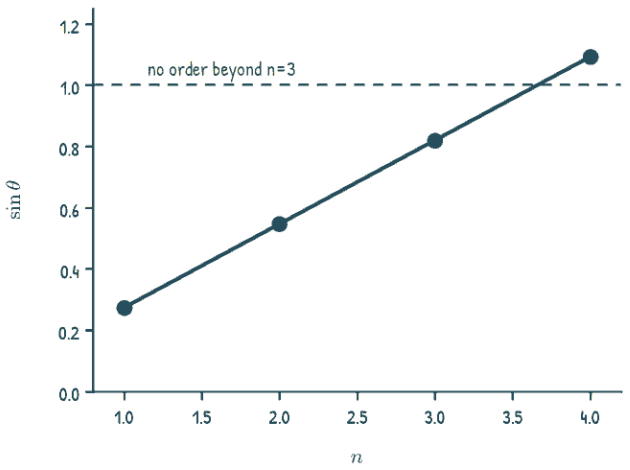
d = 282 pm, a = 564 pm

ρ = 2165 kg/m³, M = 58.44 g/mol

Bragg: path difference 2d sin(theta); orders stop when sin(theta) would exceed one




$2d\sin\theta = n\lambda$




KEEP / 2d sinθ = nλ with θ from the plane: an angle, a wavelength, and a lattice constant

# What happens when a photon hits an electron?

? Billiards with light: the shift is absolute, angle-only, and independent of everything else.

## 1 The experiment that classical physics cannot survive

Compton scattered 100 pm X-rays off graphite and found two peaks at every angle: one at the original wavelength, one shifted by an amount that grows with angle and depends on nothing else. Classical electrodynamics predicts a single unshifted line at every angle.

→ the unshifted peak is whole-atom recoil off tightly bound electrons — not an error

## 2 Conservation closes the triangle

Write energy and both momentum components, eliminate the electron angle, and use the relativistic energy-momentum relation. What falls out is a shift that depends only on the scattering angle and one constant with the dimensions of length.

$$\Delta\lambda = \frac{h}{m_e c}(1-\cos\theta) = 2.426(1-\cos\theta)\ \text{pm}$$

→ 0° → 0; 90° → 2.43 pm; 180° → 4.86 pm, the maximum

## 3 The electron takes the rest

Whatever the photon loses, the electron gains as kinetic energy. Writing x = hf/m_ec², the recoil energy has a clean closed form, and the electron always goes forward of the perpendicular — it can never come back.

$$K_e = hf\frac{x(1-\cos\theta)}{1+x(1-\cos\theta)}, \qquad \tan\phi = \frac{1}{(1+x)\tan(\theta/2)}$$

## 4 A worked back-scatter

A 100 keV photon is scattered through 180°.

$$\Delta\lambda = 4.86\ \text{pm}, \qquad x = \frac{100}{511} = 0.196$$

$$K_e = 100\frac{2(0.196)}{1+2(0.196)} = 28\ \text{keV}$$

→ the maximum energy transfer is at 180°, and it is still well below the photon energy

KEY RESULT

the shift

$$\Delta\lambda = \frac{h}{m_e c}(1-\cos\theta)$$

λ_C = 2.426 pm, a fundamental length

Absolute shift; the fraction varies with λ.

TRAP

three misreadings

Treating Δλ as a percentage of λ.

Explaining the unshifted peak as failure of the model.

Using p = E/c for the recoil electron.

NUMBERS

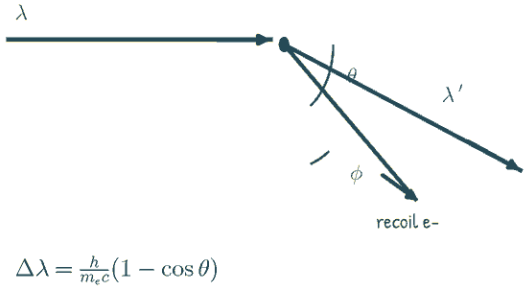
the scale

visible light: shift $10^{-5}$ of λ — invisible

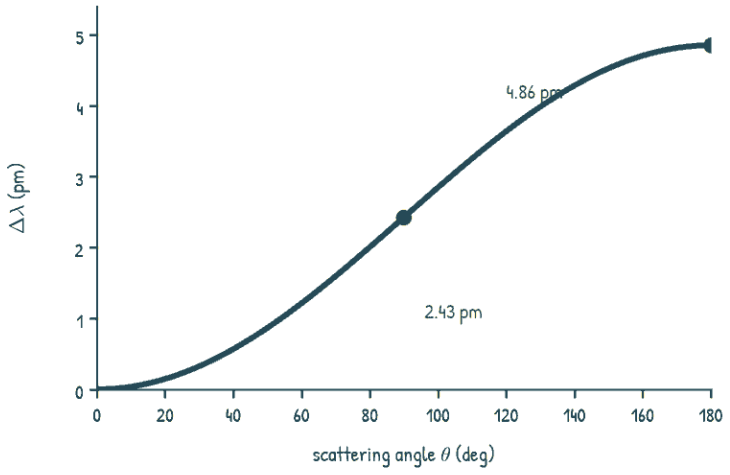
100 keV: shift is 2 % of λ

1 MeV: shift is 30 % of λ

Compton: the shift depends on angle alone - absolute, never a percentage




$$\Delta\lambda = \frac{h}{m_e c}(1-\cos\theta)$$




KEEP / $\Delta\lambda = 2.426(1-\cos\theta)$ pm: an absolute shift, never a percentage

0.1 eV 100 keV 10 MeV

# Can the Compton shift be derived without trigonometry?

? Yes — square the four-momentum of the electron and the algebra collapses.

### 1 Four-vectors make conservation linear

Energy and the three momentum components are one object. Conservation of four-momentum is then a single vector equation: the incoming photon equals the outgoing photon plus the recoil electron.

$$p_\gamma = p'_\gamma + p_e \;\Rightarrow\; p_e = p_\gamma - p'_\gamma$$

### 2 Square it and use the mass shell

The electron's four-momentum squared is $m_e^2c^2$, a number every frame agrees on. The photon's is zero. Expanding the square leaves only the dot product of the two photon momenta, which in the lab frame is a product of energies and a cosine.

$$(p_\gamma - p'_\gamma)^2 = m_e^2c^2 \;\Rightarrow\; 2p_\gamma \cdot p'_\gamma = 2m_e^2c^2$$

$$\frac{1}{\lambda'} - \frac{1}{\lambda} = \frac{1-\cos\theta}{\lambda_C}$$

→ two lines of algebra, no recoil angle, no trigonometric identities

### 3 Why this route is worth learning

The scalar derivation needs three conservation equations and one elimination. The invariant route needs one equation. When a problem has several bodies and several angles, squaring a four-momentum difference is almost always the shortest path — and it generalises to pair production and to any two-body decay.

→ same answer, but the method scales where the trigonometry does not

### 4 A worked invariant

Use the same invariant on a 2 MeV photon scattering at 60°.

$$\Delta\lambda = 2.426(1-\cos 60^\circ) = 1.21 \text{ pm}$$

$$\lambda = 0.62 \text{ pm} \;\Rightarrow\; \lambda' = 1.83 \text{ pm}$$

→ the fractional shift is 195 %: at MeV energies the photon loses most of its energy

KEY RESULT

the invariant

$$p_e = p_\gamma - p'_\gamma$$

$$p^2 = m^2c^2$$

Square first, ask questions later.

INSIGHT

why thresholds are higher

In a fixed-target collision the products must carry the target's momentum, so the centre-of-mass energy is √s, not the beam energy plus the target mass.

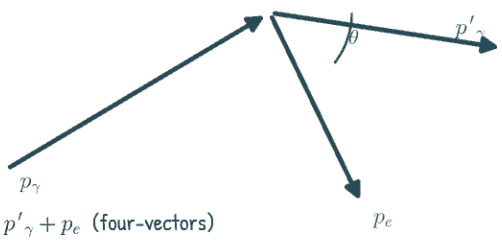
TRAP

mixing frames

The dot product p·p′ must be evaluated in one frame. Using the lab energies but a centre-of-mass angle is the standard error.

One invariant replaces three conservation equations: the Compton shift falls out in two lines

close the triangle with four-momenta




$p_\gamma = p'_\gamma + p_e$ (four-vectors)

the invariant route

square the electron line:

$$(p_\gamma - p'_\gamma)^2 = m_e^2c^2$$

$$2p \cdot p' = 2m_e^2c^2$$

$$\Rightarrow\; \Delta\lambda = \frac{h}{m_ec}(1-\cos\theta)$$

two lines, no trigonometry

KEEP / one invariant replaces three conservation equations and two eliminands

0.1 eV ▼1 keV 10 MeV

# How does a fast electron brighten a dull photon?

? The Compton event run backwards, and the reason radio galaxies shine in X-rays.

## 1 Run the collision backwards

In Compton scattering a photon hands energy to an electron. If the electron is already moving fast and the photon meets it head-on, the photon gains energy instead. In the electron's frame this is ordinary Compton scattering; transform back and the photon is boosted.

$$\epsilon' \approx \gamma^2 \epsilon \quad (\gamma\epsilon \ll m_e c^2)$$

→ a radio photon at Γ = 1000 becomes a keV photon: radio galaxies glow in X-rays

## 2 The moving-mirror picture

Treat the electron as a mirror approaching the light. One Doppler shift takes the photon into the electron frame, a second takes it back out. Two shifts of order Γ give the square.

$$\frac{\epsilon'}{\epsilon} = \frac{1+\beta}{1-\beta} \approx 4\gamma^2 \quad (\beta \to 1)$$

→ the same factor 4Γ² that appears in synchrotron and in the Doppler beaming of jets

## 3 Where it matters, and where it stops

Inverse Compton sets the X-ray output of jets, of the cosmic microwave background seen by hot cluster gas, and of any source with relativistic electrons and a soft photon bath. Once Γε approaches m_ec² the process enters the Klein-Nishina regime and the cross-section falls — that is where the simple boost stops.

→ the Sunyaev-Zel'dovich effect is this process on the microwave background

## 4 A worked boost

Microwave photons at $6\times10^{-4}$ eV hit electrons with Γ = $10^3$.

$$\epsilon' \approx 4\gamma^2\epsilon = 4(10^6)(6\times10^{-4}) = 2.4\times10^3 \text{ eV}$$

$$2.4 \text{ keV} = \text{ soft X ray}$$

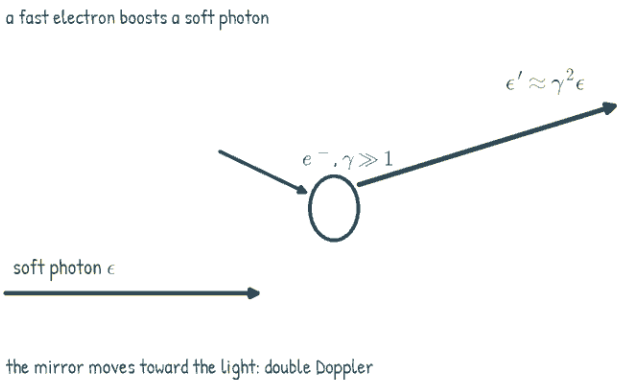
→ the same electrons synchrotron-radiate at keV too: two channels, one population

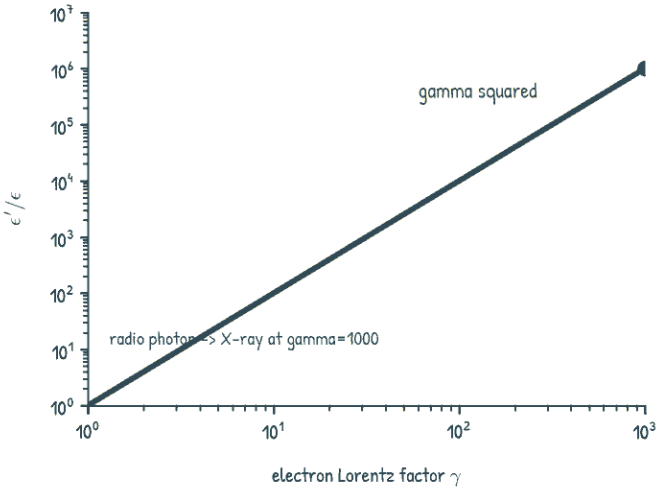
KEY RESULT
the boost
$$\epsilon' \approx \gamma^2\epsilon$$
$$\propto 4\gamma^2 \text{ at } \beta \to 1$$
two Doppler shifts, not one

GATE
when it fails
Γε ≳ m_ec²: recoil matters, the cross-section drops (Klein-Nishina).
Also assumes an isotropic soft photon bath in the electron frame.

TRAP
one shift only
Using 2Γ instead of 4Γ². The photon is boosted twice: in and out.

Inverse Compton: the same kinematics run backwards, and astrophysics uses it




KEEP / a photon bouncing off a relativistic electron gains of order Γ² in energy

# When does a photon stop being a photon?

? Above 1.022 MeV it can become matter — but never in empty space.

1

## The threshold is a rest-mass statement

Creating an electron-positron pair costs $2m_ec^2$ = 1.022 MeV of rest energy before any kinetic energy. A photon below that energy cannot do it, whatever the material.

$$E_{\min} = 2m_ec^2 = 1.022\ \text{MeV}, \qquad K_{\text{pair}} = E_\gamma - 1.022\ \text{MeV}$$

→ a 2 MeV photon leaves 0.98 MeV to share between the two particles

2

## Why empty space is forbidden

In free space, energy and momentum cannot both be conserved by one photon splitting into two equal-mass particles: in the centre-of-mass frame the photon has momentum but the pair would have none. A nucleus (or an electron) must take the recoil.

$$p_\gamma \neq p_+ + p_- \ \text{ in any frame with } E_\gamma = 2m_ec^2$$

→ the same reason a single photon cannot produce a single electron

3

## The three experiments that complete the photon

Photoelectric effect proves the energy comes in quanta. Diffraction proves the wave. Compton proves the momentum. Pair production closes the circle: at high enough energy the quantum can be converted into the very particles whose mass it never had.

→ in a bubble chamber a photon leaves no track until it converts — then two curly ones appear

4

## A worked threshold

A 2 MeV photon converts near a nucleus.

$$K_{\text{total}} = 2.00 - 1.022 = 0.98\ \text{MeV}$$

$$\text{shared as } K_+ + K_- = 0.98\ \text{MeV}$$

→ in a magnetic field the two tracks curl opposite ways with the same radius if the sharing is equal

KEY RESULT

the threshold

$$2m_ec^2 = 1.022\ \text{MeV}$$

$$K = E_\gamma - 1.022$$

needs a third body for momentum

TRAP

free-space conversion

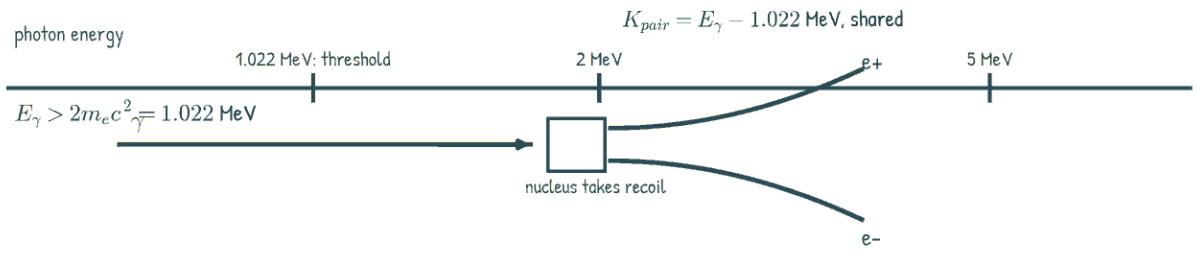
Allowing $\gamma \rightarrow e^+e^-$ with nothing else present. Momentum conservation forbids it in every frame.

INSIGHT

the photon, completed

Energy quantum (photoelectric), wave (diffraction), momentum (Compton), convertibility (pair production). Four experiments, one object.

Pair production needs 1.022 MeV and a nucleus to carry the momentum




KEEP / E > $2m_ec^2$ and a third body: energy alone is not enough, momentum must balance

# How did a straight line predict missing elements?

? Moseley plotted √f against Z in 1913 and the periodic table became a measurement.

## 1 Why the line is straight

Kα is a 2 → 1 transition in a nucleus screened by the one remaining K electron. The energy carries $(Z-1)^2$, so the frequency carries it too, and its square root is linear in Z.

$$f = \tfrac{3}{4}cR(Z-1)^2 \;\Rightarrow\; \sqrt{f} = \sqrt{\tfrac{3}{4}cR}\,(Z-1)$$

→ slope √(3cR/4) = 4.97×10⁷ (Hz)^{1/2} per unit Z

## 2 What the intercept tells you

The straight line does not pass through the origin: it hits √f = 0 at Z = 1, not Z = 0. That offset is the shielding constant, and the same trick with L lines gives b ≈ 7.4 because more electrons screen.

$$b_{K\alpha} \approx 1, \qquad b_{L\alpha} \approx 7.4$$

→ Moseley's "a and b" are exactly the slope and the shield

## 3 The prediction that found four elements

Because the line is monotonic and every element must sit on it, a gap in the sequence is a missing element rather than a gap in nature. Elements 43, 61, 72 and 75 were identified this way before they were isolated — technetium, promethium, hafnium, rhenium.

→ it also settled the ordering: Z, not atomic weight, is the periodic variable

## 4 A worked prediction

Predict the Kα wavelength of silver (Z = 47) from copper's 154 pm.

$$\lambda \propto \frac{1}{(Z-1)^2}: \;\; \lambda_{Ag} = 154\frac{28^2}{46^2} = 57 \text{ pm}$$

$$f = \tfrac{3}{4}cR(46)^2 = 5.3\times10^{18} \text{ Hz}$$

→ the measured value is 56 pm: one line, one constant, every element

KEY RESULT

Moseley

$$\sqrt{f} = a(Z-b)$$

Kα: b ≈ 1; Lα: b ≈ 7.4

Z is the ordering variable, not mass.

INSIGHT

what it changed

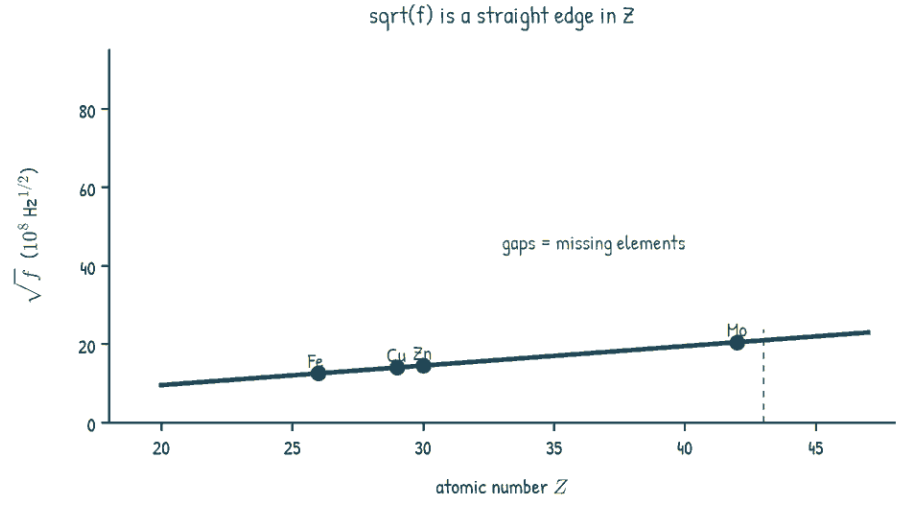
Before 1913 the periodic table was a chemical pattern with guessed gaps. After it, the gaps were places with a known X-ray address.

TRAP

one line, one constant

Using a Kα value of b for an L line, or treating b as universal.

Moseley: the K-alpha frequency is a ruler for Z, with the missing elements as gaps




$$E_{K\alpha} = \tfrac{3}{4}(13.6)(Z-1)^2$$

one K electron

screens the nucleus

L lines use

$b \approx 7.4$

so the constant is line-specific

KEEP / √f ∝ (Z − b): every element is a point on one line, and gaps mean missing elements

▼13.6 eV
0.1 eV 10 MeV

# How long would a classical atom live?

? Sixteen picoseconds, and a continuous spectrum while it dies. That is the crisis.

## 1 An accelerating charge radiates

Larmor's formula gives the power radiated by an accelerating charge. For an electron in a circular orbit the acceleration is $v^2/r$, so the power climbs steeply as the orbit shrinks — which makes the collapse accelerate as it proceeds.

$$P = \frac{e^2 a^2}{6\pi\varepsilon_0 c^3}, \qquad a = \frac{ke^2}{m_e r^2}$$

## 2 Integrate the energy loss

The orbital energy is $-ke^2/2r$, so losing energy means losing radius. Substituting gives a separable equation whose integral is a lifetime proportional to the cube of the starting radius.

$$\frac{dE}{dt} = -P \;\Rightarrow\; r^2\dot{r} = -\frac{2C}{ke^2} \;\Rightarrow\; \tau = \frac{ke^2 r_0^3}{6C}$$

$$\tau(a_0) = 1.6 \times 10^{-11}\ \text{s}$$

→ $\tau_C \approx 5$ mm: a slow spiral of ten million orbits, not a plunge

## 3 Two predictions that both fail

The atom should live 16 ps, and while dying its orbital frequency should sweep continuously upward, emitting a continuum from infrared to X-rays. Real atoms live forever and emit sharp lines. Both failures are fixed by the same assumption: stationary states.

→ the stability and the line spectrum are one problem, not two

## 4 A worked collapse time

Evaluate the Larmor-integral lifetime from $a_0$:

$$C = \frac{e^2}{6\pi\varepsilon_0 c^3}\left(\frac{ke^2}{m_e}\right)^2 = 6.3 \times 10^{-55}\ \text{J m}^4$$

$$\tau = \frac{ke^2 a_0^3}{6C} = 1.6 \times 10^{-11}\ \text{s}$$

→ $\tau_C \approx 5$ mm and about $10^9$ orbits: slow, and fatal

KEY RESULT
the collapse

$$P = e^2 a^2 / 6\pi\varepsilon_0 c^3$$

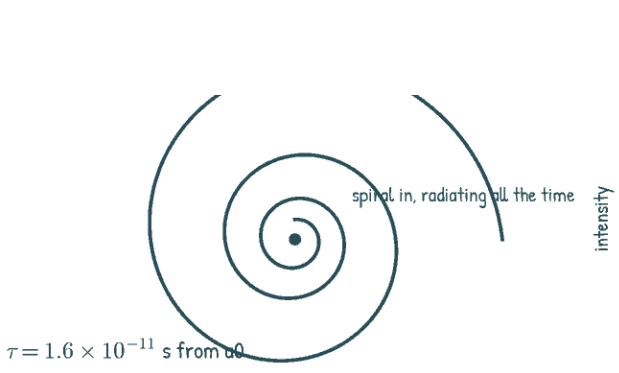
$$\tau = ke^2 r_0^3 / 6C$$

16 ps from $a_0$; a continuum, not lines

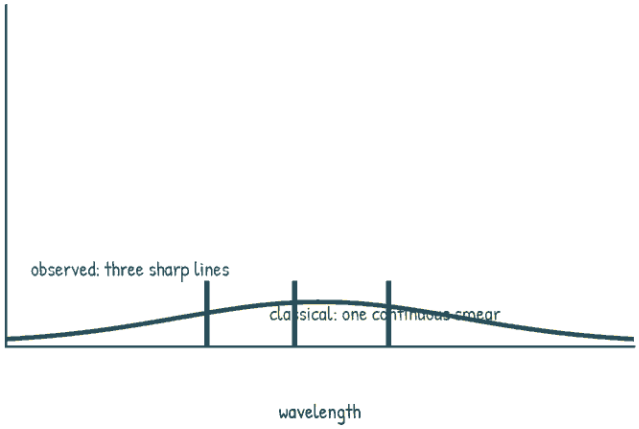
INSIGHT
why it is a spiral
$\tau_C \approx 5$ mm is five orders larger than the atom, so the electron completes ~$10^9$ orbits while dying. The "instant collapse" story is wrong in scale.

TRAP
the wrong lifetime
Quoting $10^{-8}$ s — that is an excited-state lifetime, not the classical collapse. The collapse is $10^{-11}$ s.

The classical atom lives 16 ps and glows a continuum; real atoms do neither




a dying atom cannot be line-bright




KEEP / $\tau \propto r^3$: the collapse is a slow spiral, not a plunge, and it radiates every frequency




# Why does quantum mechanics become classical for large n?

? Expand the level spacing and the photon frequency turns into the orbital frequency.

1

## A large orbit is a classical antenna

An electron on a big orbit radiates at its orbital frequency. Quantum mechanics says it radiates at the energy gap divided by h. For the two to agree, the gap must approach the orbital energy spacing — which is what a Taylor expansion shows.

$$\Delta E = \frac{A}{n^2}\left[(1-1/n)^{-2}-1\right] \approx \frac{2A}{n^3}\left(1+\frac{3}{2n}\right)$$

$$f_{\text{photon}} = f_{\text{orbit}}\left(1+\frac{3}{2n}\right)$$

2

## Check the numbers at n = 100

The correction is 1.5 %. At n = 10 it is 15 %, and the quantum world is visibly non-classical. The ratio approaches one from above, monotonically, and the limit is the statement that quantum mechanics contains classical mechanics.

$$n=100:\ \ f_{\text{orb}} = 6.58\times10^{9}\ \text{Hz},\ \ f_{\text{ph}} = 6.68\times10^{9}\ \text{Hz}$$

→ Rydberg atoms at n ~ 100 are used as sensors precisely because they are almost classical

3

## Why this is more than a curiosity

Correspondence is the consistency condition any new theory must satisfy: reduce to the old one where it works. Bohr used it to guess selection rules before the full theory existed.

4

## A worked n = 100 check

Compare orbital and photon frequency at n = 100:

$$f_{\text{orb}} = \frac{v}{2\pi r} = 6.58\times10^{9}\ \text{Hz}$$

$$f_{\text{ph}} = \frac{\Delta E}{h} = 6.68\times10^{9}\ \text{Hz}$$

→ ratio 1.015, exactly 1 + 3/(2n) with n = 100

KEY RESULT

the limit

$$\Delta E/h = f_{\text{orb}}(1+3/2n)$$

Correction falls as 1/n

Consistency condition, not a coincidence

GATE

what it does not say

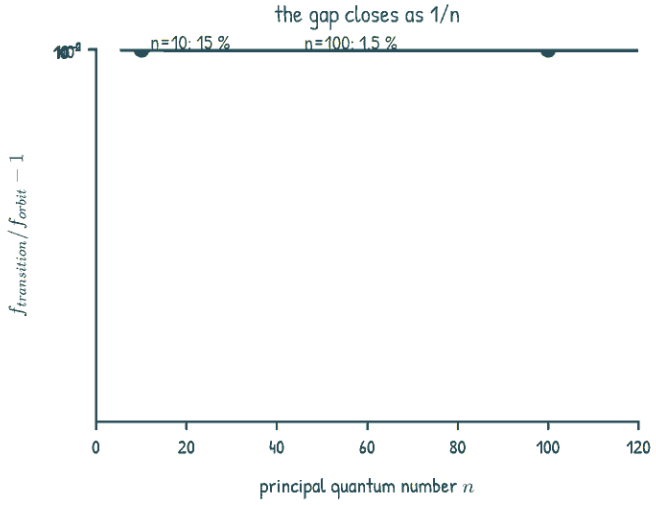
It does not say the classical trajectory is recovered for every state — only for coherent, large-n ones.

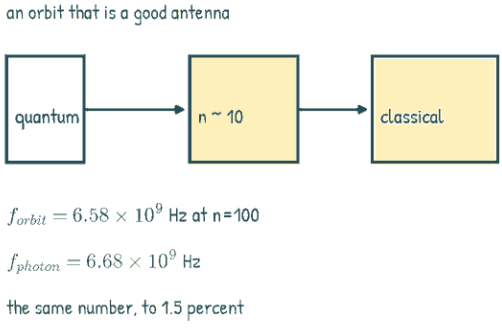
NUMBERS

bench

n = 10: 15 % · n = 100: 1.5 % · n = 150: 1 %

Correspondence: for large n the photon frequency melts into the orbital frequency




KEEP / for large n the gap closes as 1/n, and the photon frequency equals the orbit frequency to 1.5 %

# What if the electron weighed 207 times more?

? Muonic hydrogen is a nuclear ruler; positronium is a hydrogen twice as big.

1

## The substitution is the whole physics

The two-body problem reduces to one body of reduced mass μ orbiting a fixed centre. Every result of the Bohr derivation then goes through unchanged with m_e replaced by μ, so the radius scales as 1/μ and the binding as μ.

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \qquad a \propto \frac{1}{\mu}, \qquad E \propto \mu$$

2

## Muonic hydrogen: an atom the size of a nucleus

A muon is 207 times the electron mass, so the ground orbit is 256 fm — inside the nuclear charge distribution of a proton. The levels therefore shift by a measurable fraction, and comparing them with the point-nucleus prediction measures the proton radius.

$$a_\mu = \frac{0.529\ \text{Å}}{207} = 256\ \text{fm}, \qquad E_1 = -2.81\ \text{keV}$$

→ this is the "proton radius puzzle": muonic and electronic measurements disagreed for a decade

3

## Positronium: half the mass, twice the size

Two equal masses give μ = m/2, so the "atom" is twice as big and half as bound, and every wavelength doubles. It also annihilates — the ground state lives 125 ps — which makes it a laboratory for QED rather than for chemistry.

$$a = 1.06\ \text{Å}, \qquad E_1 = -6.8\ \text{eV}, \qquad \lambda_{2\to1} = 243\ \text{nm}$$

4

## A worked positronium line

The 2 → 1 line of positronium:

$$\Delta E = 6.8\left(\tfrac{3}{4}\right) = 5.1\ \text{eV}$$

$$\lambda = \frac{1240}{5.1} = 243\ \text{nm}$$

→ exactly twice hydrogen's 121.6 nm, because μ is exactly half

KEY RESULT
exotic scaling

$$\mu = m_1 m_2/(m_1 + m_2)$$

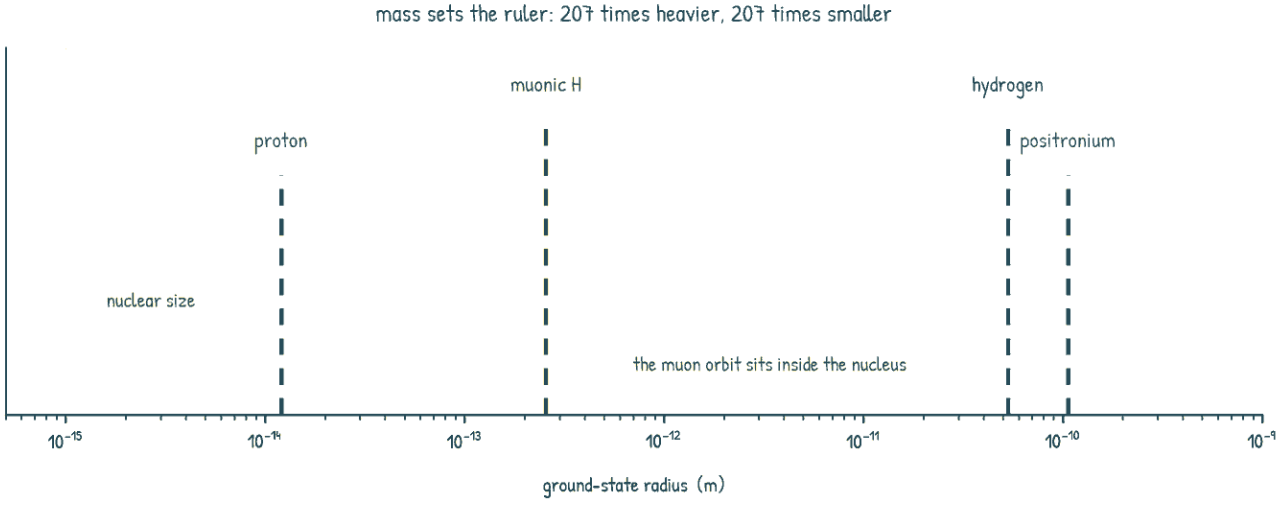
$$a \propto 1/\mu, \quad E \propto \mu$$

E·a is invariant.

INSIGHT
why muonic atoms measure radii
The wavefunction overlaps the nucleus, so the level depends on the charge distribution rather than on a point charge.

TRAP
using m_e
Computing muonic hydrogen with the electron mass gives an answer wrong by 207 — and the proton-radius claim wrong by 207 too.

Exotic atoms are scaling tests: replace the electron mass by the reduced mass




KEEP / every mass-dependent result takes μ: 207 times heavier, 207 times smaller

# Can you see energy levels without looking at light?

? Yes: fire electrons at atoms and watch the current develop teeth.

1

## The experiment in one sentence

Electrons accelerate through mercury vapour toward a grid. If the vapour had no internal structure they would gain energy continuously and the current would rise smoothly. Instead it dips periodically: at certain energies an electron can hand 4.9 eV to an atom and keep only the remainder, so it no longer has enough to reach the collector.

→ elastic collisions dominate below 4.9 eV; inelastic ones switch on exactly at it

2

## Why the dips repeat

After an inelastic collision the electron restarts from near rest. It therefore needs another 4.9 eV of free path before it can excite again, so the collector current dips at every multiple of the threshold voltage.

$$V_{\text{dip}} = n\,(4.9\ \text{V}), \qquad n = 1, 2, 3$$

→ the spacing, not the position, is the measurement: it is insensitive to contact potentials

3

## What it proves that spectroscopy does not

Optical spectroscopy sees the photons that come out; Franck-Hertz sees the energy that goes in. Together they close the book: the levels are real, they are discrete, and they are the same levels whether you excite them with electrons or with light. It also shows that excitation costs a quantum, not a continuous trickle.

→ the mercury atom then emits 254 nm ultraviolet as it de-excites — the same 4.9 eV

4

## A worked dip position

Where do the first three dips sit, and why not at exactly 4.9 V?

$$V_{\text{dip}} \approx V_{\text{contact}} + n(4.9\ \text{V}), \quad n = 1, 2, 3$$

$$4.9,\ 9.8,\ 14.7\ \text{V plus a constant offset}$$

→ the spacing is the measurement; the offset is contact potential and is subtracted, not explained

KEY RESULT

the measurement

$$\Delta V = 4.9\ \text{V}\ (\text{Hg})$$

dips at $n\Delta V$

inelastic threshold = level spacing

GATE

why the vapour pressure matters

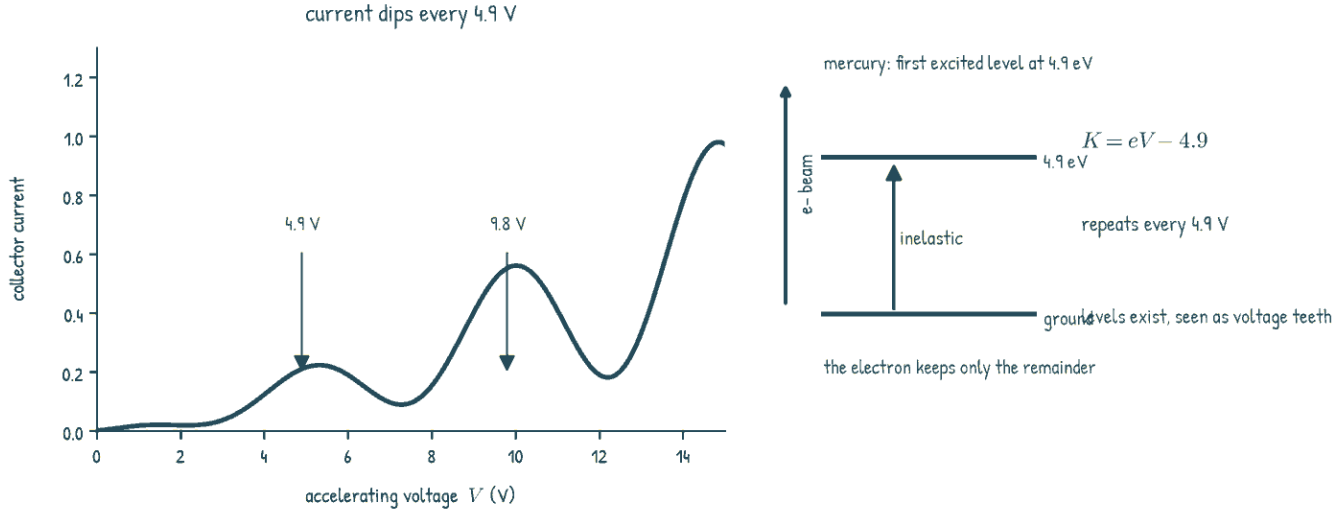
Too little vapour and the electrons never collide; too much and they scatter many times before the grid. The window is narrow, which is why the original experiment was delicate.

TRAP

reading the first dip

The first dip is shifted by the contact potential difference between the electrodes. Only the spacing is reliable.

Franck-Hertz turns a level diagram into a current-voltage curve with teeth at 4.9 V




KEEP / a dip every 4.9 V in mercury is a level diagram measured with an ammeter

# How long should a wave take to eject an electron?

? Give classical physics its best case, then compare with the clock.

1

## Give the wave picture every advantage

Assume the light is spread evenly, that an atom absorbs perfectly over its own cross-section, and that nothing is lost. Then the time to accumulate the escape energy is the energy needed divided by the power intercepted.

$$t = \frac{\phi}{IA} = \frac{3.2\times10^{-19}}{(10^{-3})(10^{-20})} = 3.2\times10^{4}\ \text{s}$$

→ about nine hours at a milliwatt per square metre — and a century at $10^{-8}$ W/m²

2

## The clock says otherwise

The measured delay between switching on the light and the first electrons is below a nanosecond at every intensity tested, including beams so dim that the wave estimate runs to hours. No accumulation time exists.

$$t_{\text{observed}} \lesssim 10^{-9}\ \text{s} \quad \text{vs} \quad t_{\text{wave}} \sim 10^{4}\ \text{s}$$

→ the ratio is $10^{13}$; this is not a discrepancy of factors of two

3

## Why the argument is decisive rather than rhetorical

The estimate is deliberately generous. Even if the atom's cross-section were a hundred times larger the wait would be minutes, and even at sunlight intensity it would be milliseconds. Any single-photon process removes the difficulty: the energy arrives concentrated in one quantum, so there is nothing to accumulate.

→ the only escape for the wave picture is a collective absorption mechanism, which nobody has ever observed

4

## A worked dim-beam estimate

Moonlight on one atom: I = $10^{-3}$ W/m², A = $10^{-20}$ m², Φ = 2 eV.

$$P = IA = 10^{-23}\ \text{W}, \qquad t = \frac{3.2\times10^{-19}}{10^{-23}} = 3.2\times10^{4}\ \text{s}$$

$$\approx 9\ \text{hours}, \quad \text{observed} < 10^{-9}\ \text{s}$$

→ thirteen orders of magnitude: the wave picture is not wrong by a factor, it is wrong

KEY RESULT

the comparison

$$t = \phi/IA$$

$10^{4}$ s predicted, $10^{-9}$ s observed

one photon, one electron removes the problem

INSIGHT

why thresholds follow

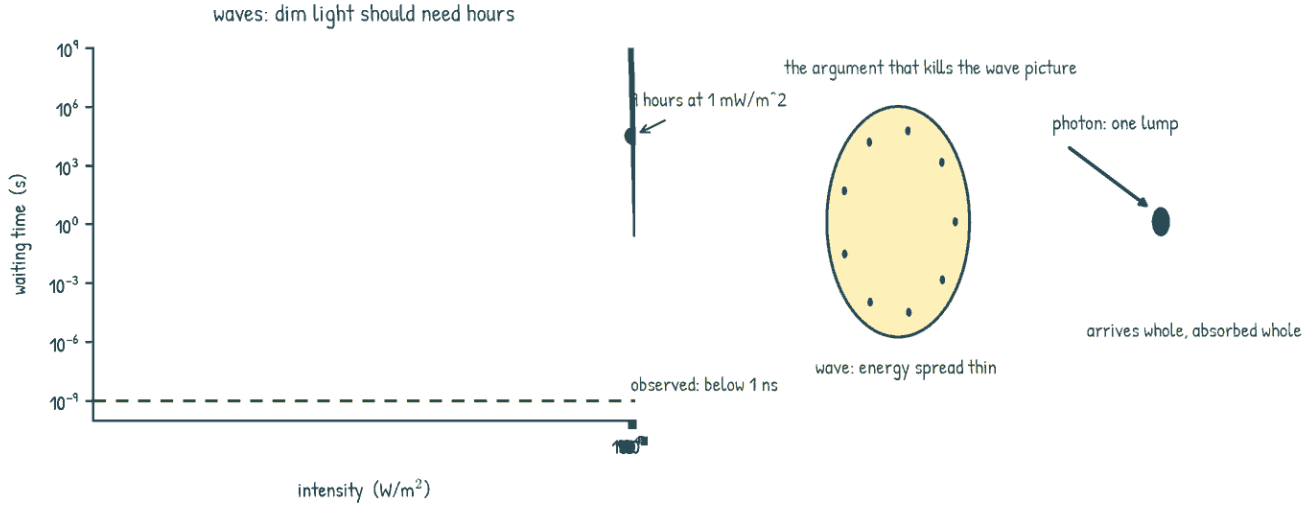
If energy arrives in quanta, then a quantum smaller than the toll simply cannot pay it — however many arrive. That is the threshold, and it needs no extra assumption.

TRAP

intensity vs energy

Saying "more intense light has more energy per photon". Intensity is photons per area per second; the energy per photon is fixed by frequency.

Classical waiting time at dim light is hours; the observed delay is nanoseconds




KEEP / hours at moonlight intensity versus nanoseconds observed: thirteen orders of magnitude

0.1 eV 511 keV 10 MeV

# Which speed does a matter wave travel at?

? Two speeds, one particle, and a product that is always c².

### 1 Two velocities from one dispersion relation

A wave has a phase velocity ω/k and a group velocity dω/dk. For a matter wave the first is the speed of the crests and the second the speed of the envelope — the packet that actually represents the particle.

$$v_p = \frac{\omega}{k} = \frac{E}{p}, \qquad v_g = \frac{d\omega}{dk} = \frac{dE}{dp}$$

### 2 Substitute the relativistic dispersion relation

With $E^2 = p^2c^2 + m^2c^4$, differentiating gives $dE/dp = pc^2/E$. Dividing E by p gives the reciprocal, so the two speeds multiply to c². The group velocity comes out exactly the particle speed.

$$v_p = \frac{c^2}{v} > c, \qquad v_g = v, \qquad v_p v_g = c^2$$

→ a single infinite sine carries no information, so v_p > c violates nothing

### 3 Why this is not a loophole

Signal speed is the speed of a change in the envelope, which is the group velocity. For a free particle that is v < c. For light in a plasma the phase velocity also exceeds c while the group velocity is below it — the same structure, and the same reason no message outruns light.

→ a wave packet with sharp edges spreads; its front still moves at v_g

### 4 A worked c² product

An electron moving at 0.6c:

$$v_p = \frac{c^2}{0.6c} = 1.67c, \qquad v_g = 0.6c$$

$$v_p v_g = c^2 \ \checkmark$$

→ the crests outrun light while the particle does not; only the packet carries information

KEY RESULT
two speeds

$$v_p = E/p = c^2/v$$

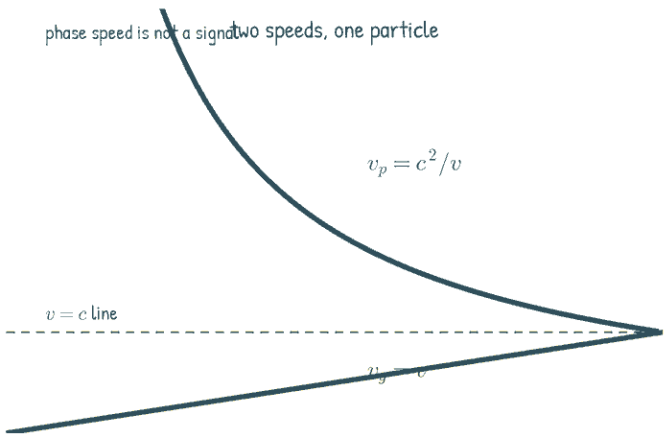
$$v_g = dE/dp = v$$

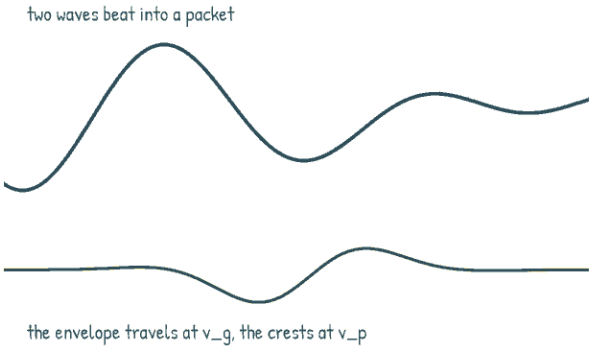
$$v_p v_g = c^2$$

GATE
when v_p > c is fine
Phase velocity is a property of a mathematical surface of constant phase, not of a physical object. No energy, no information.

TRAP
the photon limit
For m → 0 both speeds tend to c, and the distinction collapses. Do not carry the formula past that limit as if v_p were meaningful.

Matter wave: v_p v_g = c squared; only the group speed carries the particle




KEEP / phase speed c²/v exceeds c harmlessly; group speed v carries everything

0.1 eV — 1 MeV ▼ — 10 MeV

# Why does a factor of two in energy swing a half-life by $10^{24}$?

? Alpha decay spans thirty orders of magnitude in lifetime, yet the energies differ by a factor of two.

### 1 A barrier the alpha cannot climb

An alpha inside a heavy nucleus faces a Coulomb hill tens of MeV high while carrying only 4–9 MeV: classically it never leaves. It tunnels, with an exponent set by barrier width and height.

$$\log_{10}\lambda \approx a - \frac{bZ}{\sqrt{E}}$$

→ E in MeV, and the second term is of order 40 — an exponent, not a correction

### 2 Two nuclides show the amplifier

U-238 emits 4.27 MeV alphas and lives $4.5\times10^{9}$ years; Po-212 emits 8.95 MeV alphas and lives 0.3 µs. The argument $1/\sqrt{E}$ changes by 30 %, and the lifetime changes by twenty-four decades.

$$\frac{1}{\sqrt{4.27}} = 0.484, \qquad \frac{1}{\sqrt{8.95}} = 0.334\ \text{MeV}^{-1/2}$$

→ a 30 % change in the argument, a $10^{24}$ change in the answer: exponential amplification

### 3 The straight line, and what it is good for

For isotopes of one element, log T against $1/\sqrt{E}$ is the Geiger–Nuttall line. Its slope grows with Z, so an unknown alpha energy lands on one element's line and not another's.

$$\log_{10} T_{1/2} = A + \frac{BZ}{\sqrt{E}}$$

→ Gamow explained in 1928 what Geiger and Nuttall had measured in 1911

### 4 A worked decay constant

Turn the two half-lives into rate constants, then take the ratio.

$$\lambda_{238} = \frac{\ln 2}{4.5\times10^{9}(3.156\times10^{7})} = 4.9\times10^{-18}\ \text{s}^{-1}$$

$$\lambda_{212} = \frac{\ln 2}{3\times10^{-7}} = 2.3\times10^{6}\ \text{s}^{-1}$$

→ a ratio of $4.7\times10^{23}$ in rate for a factor 2.1 in alpha energy

KEY RESULT
the law

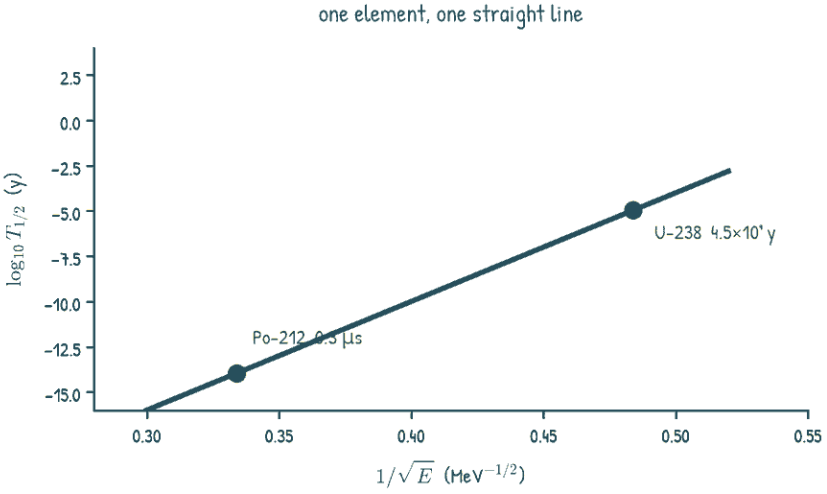
$$\log\lambda = a - bZ/\sqrt{E}$$

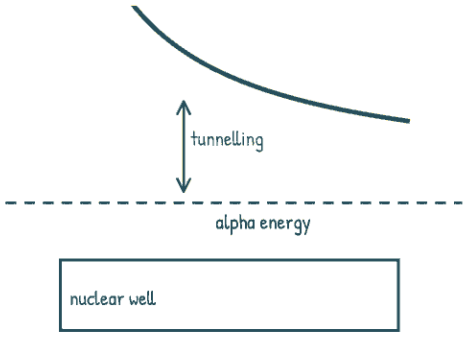
Same-Z isotopes: one straight line.
Slope ∝ Z.

GATE
what it ignores
Prefactors from nuclear structure, angular momentum hindrance, and deformation.
It is a systematics law, not a rate calculation.

TRAP
the axis
Plotting log T against E rather than $1/\sqrt{E}$ destroys the line — and the physics is in that choice of axis.

log T is linear in $1/\sqrt{E}$: a 30 % change in the argument moves the lifetime by 24 decades




KEEP / log T is linear in $1/\sqrt{E}$: the tunnelling exponent is the amplifier

# Why does U-235 fission with slow neutrons and U-238 not?

? Same element, one neutron apart, completely different behaviour. The answer is a pairing bonus of 0.8 MeV.

1

## Deforming the drop costs surface and saves Coulomb

Stretch the drop and its surface area grows (a cost) while the protons move apart (a gain). The barrier is the hump between: the drop must be pushed over it before Coulomb can run away.

$$B_f \approx 4\pi R^2 \gamma\left(1 - \frac{Z^2}{2A}\right)$$

→ the barrier falls as the fissility $Z^2/A$ grows; it vanishes near $Z^2/A \approx 50$

2

## Fissility is 36.0 for U-235 and 35.6 for U-238

Both sit far below the limit, so neither fissions on its own. What decides is the excitation energy supplied by the captured neutron, compared with the barrier height.

$$\frac{Z^2}{A} = \frac{92^2}{235} = 36.0, \qquad \frac{92^2}{238} = 35.6$$

→ the difference is small; the pairing term is what actually flips the outcome

3

## One pairing term, two destinies

Adding a neutron to U-235 makes even-even U-236, which collects a pairing bonus of about $12/\sqrt{236} \approx 0.8$ MeV on top of the neutron's binding. U-238 plus a neutron makes odd-even U-239 with no bonus, so only a fast neutron's kinetic energy closes the gap.

$$E^*_{236} \approx 6.5\ \text{MeV} > B_f, \qquad E^*_{239} \approx 4.8\ \text{MeV} < B_f$$

→ thermal neutrons for U-235, fast neutrons for U-238 — one line of liquid-drop arithmetic

4

## A worked pairing bonus

The excitation of U-236 against its barrier.

$$\delta = \frac{12}{\sqrt{236}} = 0.78\ \text{MeV}$$

$$E^*_{236} = 5.7 + 0.78 = 6.5\ \text{MeV} > B_f \approx 6\ \text{MeV}$$

→ U-239 gets 4.8 MeV and stays below the barrier: no pairing bonus, no thermal fission

KEY RESULT

the numbers

Barrier ≈ 6 MeV for both.

Excitation 6.5 vs 4.8 MeV.

Fissility 36.0 vs 35.6.

INSIGHT

why even-even pays

Nucleons pair with opposite spins; an unpaired nucleon costs energy. Even-even nuclei are the tightly bound ones.

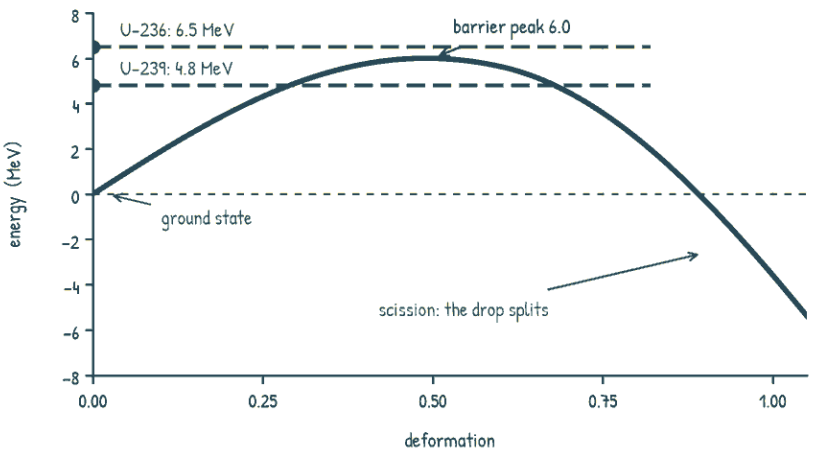
TRAP

the fissility shortcut

Using $Z^2/A$ alone to predict which isotope fissions. It sets the barrier's trend, not the outcome for one neutron.

Deformation costs surface and saves Coulomb; pairing decides which nucleus gets over the hump

one barrier; the excitation decides who clears it




surface cost, then Coulomb gain




KEEP / the barrier is ~6 MeV for both; pairing decides who clears it

# Can you derive the valley rather than remember it?

? Maximise the liquid-drop binding over charge at fixed mass. One derivative gives the stability line.

1 **Only two terms care about Z**

At fixed A the volume and surface terms are constants, and pairing is a small step. The competition is between the Coulomb term (grows as $Z^2$) and the asymmetry term (grows as $(A-2Z)^2$).

$$\frac{\partial B}{\partial Z} = -2a_c\frac{Z}{A^{1/3}} - 2a_a\frac{2(A-2Z)}{A} = 0$$

→ the asymmetry term is written so that N − Z = A − 2Z

2 **Solve it**

Rearranging gives the floor explicitly. Small A returns Z = A/2; large A pushes the floor toward neutron-rich nuclei because the Coulomb tax grows faster than the asymmetry tax.

$$Z_0 = \frac{4a_a A}{8a_a + a_c A^{2/3}} = \frac{A}{2 + \frac{a_c}{2a_a}A^{2/3}}$$

→ a_c/2a_a = 0.71/46.4 = 0.0153, the 0.0154 of the formula sheets

3 **Test it where the data are**

At A = 100 the formula gives 42.9; the stable isobars are Mo and Ru, Z = 42–44. At A = 238 it gives 92.0 and uranium is at 92. The formula is a mean trend: shell closures dent it, which is why the real chart has kinks.

$$Z_0(100) = 42.9, \qquad Z_0(238) = 92.0$$

→ the liquid drop is the climate; the shells are the weather

4 **A worked coefficient**

Check the 0.0154 in the valley formula.

$$\frac{a_c}{2a_a} = \frac{0.71}{2(23.2)} = 0.0153$$

$$Z_0(56) = \frac{56}{2 + 0.0153(56)^{2/3}} = 27.0$$

→ iron sits at Z = 26: the liquid drop is one unit high, and the shell closure explains it

KEY RESULT

the derivative

$$Z_0 = \frac{A}{2 + 0.0154A^{2/3}}$$

N = Z at small A.

N/Z → 1.5 at A = 238.

GATE

what is dropped

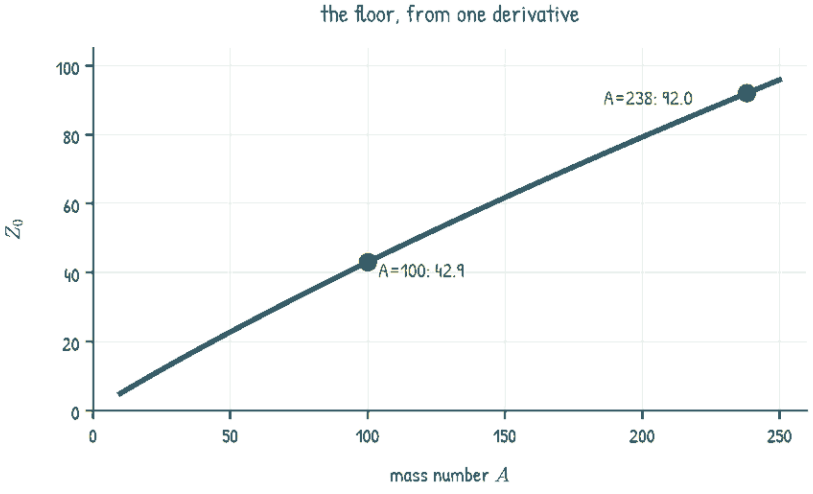
Pairing's derivative is ignored and shell effects are absent, so the formula misses magic-number nuclei by several units of Z.

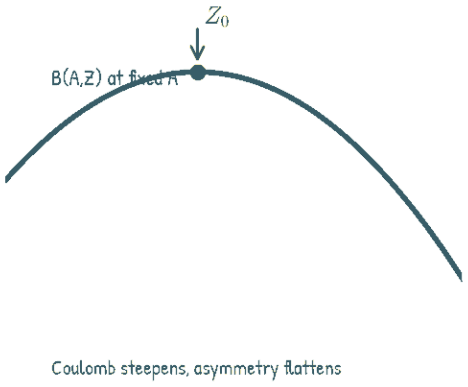
TRAP

differentiating A

Treating A as a variable. The valley is defined at fixed A — that is the whole point of an isobar.

Differentiate B over Z at fixed A: the valley floor follows, and shells dent it




KEEP / differentiate B(A,Z): Z₀ = A/(2 + (a_c/2a_a)A^{2/3})

0.1 eV 1 MeV 10 MeV

# How long does a daughter take to peak?

? Solve the two-step chain properly, then read the equilibrium and the peak time off the solution.

### 1 Solve the pair with an empty daughter

Substitute the parent's exponential into the daughter's equation and integrate with $N_2(0) = 0$. The result is a difference of two exponentials: the daughter's own decay minus the parent's supply.

$$N_2(t) = \frac{\lambda_1 N_1(0)}{\lambda_2 - \lambda_1}\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right)$$

→ when $\lambda_2 > \lambda_1$ the second exponential dies first and the parent survives

### 2 Late times leave one exponential

After the daughter's own transient has died away, the bracket approaches e^{-λ₁t} and the ratio of activities becomes a constant. That constant is the equilibrium, and it is 1 in the secular limit.

$$\frac{A_2}{A_1} \to \frac{\lambda_2}{\lambda_2 - \lambda_1} \to 1 \quad \text{as } \lambda_1 \ll \lambda_2$$

→ transient ratio 1.5 when $\lambda_2 = 3\lambda_1$; secular ratio 1 when the parent is very long-lived

### 3 Differentiate again for the peak

Setting the derivative of $N_2$ to zero gives the time of maximum daughter activity. It is the crossing of the two exponentials — the moment the daughter's production and decay balance at their largest.

$$t_{\max} = \frac{\ln(\lambda_2/\lambda_1)}{\lambda_2 - \lambda_1}$$

→ the peak matters for a generator: elute the daughter at the maximum, not later

### 4 A worked maximum

$\lambda_1 = 0.02\ \text{h}^{-1}$, $\lambda_2 = 0.10\ \text{h}^{-1}$: ratio and peak time.

$$\frac{A_2}{A_1} = \frac{0.10}{0.08} = 1.25$$

$$t_{\max} = \frac{\ln 5}{0.08} = 20.1\ \text{h}$$

→ the daughter peaks at 20 h, then falls with the parent's 34.7 h half-life

KEY RESULT

the three results

$$N_2(t) \propto e^{-\lambda_1 t} - e^{-\lambda_2 t}$$

$$A_2/A_1 = \lambda_2/(\lambda_2 - \lambda_1)$$

$$t_{\max} = \ln(\lambda_2/\lambda_1)/(\lambda_2 - \lambda_1)$$

GATE

limits behave

$t \to 0$: $N_2 \to 0$ with zero slope.

$\lambda_1 \to 0$: ratio → 1, secular.

Dimensions: t_max is a time.

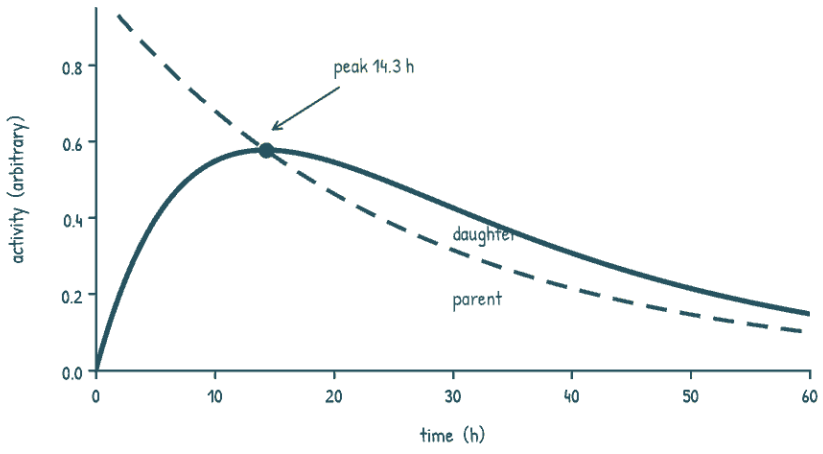
TRAP

which exponential

Subtracting the parent from the daughter (sign flip) gives a negative activity. The daughter term always decays faster.

$A_2/A_1$ settles at $\lambda_2/(\lambda_2 - \lambda_1)$; the peak is where production and decay balance

rise, peak, then follow the parent




equilibrium ratio

transient: 1.5

secular: 1.0

the longer-lived parent sets the tempo

KEEP / $A_2$ peaks at $\ln(\lambda_2/\lambda_1)/(\lambda_2 - \lambda_1)$, then follows the parent

# How many solar neutrinos cross your thumbnail each second?

? The Sun's luminosity fixes its fusion rate, and the fusion rate fixes the neutrino flux at Earth.

KEY RESULT
the chain of division
L → helium rate → neutrino rate → flux at 1 AU.
26.7 MeV and two neutrinos per helium.

GATE
the assumptions
Steady Sun, p-p chain dominant (CNO is a few per cent), and no neutrino loss inside the Sun.

TRAP
one neutrino or two
Counting one neutrino per helium halves the flux; the chain emits two, one per positron annihilation step.

### 1 Luminosity counts helium nuclei

Every 26.7 MeV released corresponds to one helium nucleus made and two neutrinos emitted. Dividing the luminosity by that energy gives the helium rate directly, with no model of the core needed.

$$\dot{N}_{\mathrm{He}} = \frac{3.83\times10^{26}}{26.7\times1.602\times10^{-13}} = 8.9\times10^{37}\ \mathrm{s}^{-1}$$

→ twice that many neutrinos: the chain emits two per helium

### 2 Spread them over a sphere

At Earth the flux is that rate divided by 4π times the astronomical unit squared. The number is enormous, and it is the number every solar-neutrino experiment had to explain.

$$\phi_\nu = \frac{1.8\times10^{38}}{4\pi(1.496\times10^{11})^2} = 6.4\times10^{14}\ \mathrm{m}^{-2}\mathrm{s}^{-1}$$

→ through one square centimetre: 6×10¹⁰ per second, through a thumbnail every second

### 3 The detected flux is a third of this

The measured flux on Earth is about a third of the prediction, and the resolution is neutrino oscillation: electron neutrinos convert into muon and tau flavours on the way out. The total flux is right; only the flavour mix changes.

$$\phi_{\nu_e} \approx \tfrac{1}{3}\phi_\nu \quad \text{(oscillated)}, \qquad \phi_{\text{total}}\ \text{conserved}$$

→ a flux measurement is a flavour measurement, not a luminosity test

### 4 A worked flux

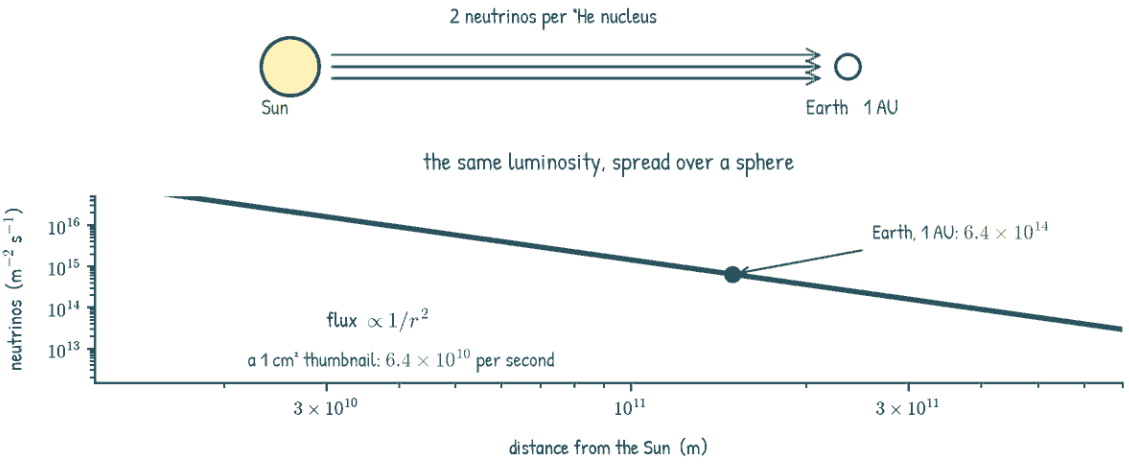
Helium rate, neutrino rate, and the flux at 1 AU.

$$\dot{N}_{\mathrm{He}} = \frac{3.83\times10^{26}}{26.7(1.602\times10^{-13})} = 8.9\times10^{37}\ \mathrm{s}^{-1}$$

$$\phi = \frac{2(8.9\times10^{37})}{4\pi(1.496\times10^{11})^2} = 6.4\times10^{14}\ \mathrm{m}^{-2}\mathrm{s}^{-1}$$

→ 6×10¹⁰ per second through a square centimetre, day and night

Luminosity counts helium nuclei; helium counts neutrinos; geometry turns that into a flux




KEEP / L/26.7 MeV gives 1.8×10³⁸ neutrinos per second; 6×10¹⁴ m⁻²s⁻¹ at Earth

# How much fuel does a reactor eat in a day?

? A gigawatt is a mass rate in disguise, and ten half-lives is an engineering unit of forgetfulness.

1

## Energy per day, then mass per day

A gigawatt thermal for a day is 8.6×10¹³ J. Dividing by the energy released per gram of U-235 gives about a kilogram — the whole fuel bill of a large reactor fits in a briefcase.

$$E_{\text{day}} = 10^9(86400) = 8.6\times10^{13}\ \text{J}, \qquad m = \frac{8.6\times10^{13}}{8.2\times10^{10}} = 1.05\ \text{kg}$$

→ at 33 % efficiency an electrical gigawatt needs three times that, about 3.2 kg/day

2

## Why a kilogram is not a gram

Only the U-235 fissions; natural uranium is 0.7 % U-235, and a reactor enriches it to a few per cent. The mass of uranium consumed is therefore tens of times the mass fissioned, and most of the spent fuel is still uranium.

$$m_{\text{U}} \approx \frac{1.05}{0.04} \approx 26\ \text{kg/day}\quad \text{fed}$$

→ at 4 % enrichment; the difference between burned and fed is the fuel cycle

3

## Ten half-lives is the storage shorthand

Each half-life removes a factor two; ten of them remove a factor 1024. That is why "store it for ten half-lives" is the phrase in waste policy: it turns an alarming activity into a small fraction, without ever reaching zero.

$$2^{-10} = 9.8\times10^{-4} \approx 0.1\,\%$$

→ a 30-year isotope stored 300 years is ten half-lives — 0.1 % of the original activity

4

## A worked fuel bill

1 GW thermal for one day, and the activity after ten half-lives.

$$m = \frac{10^9(86400)}{8.2\times10^{10}} = 1.05\ \text{kg}$$

$$2^{-10} = 9.8\times10^{-4}$$

→ a kilogram a day, and a thousandfold drop per ten half-lives — both are rules of thumb

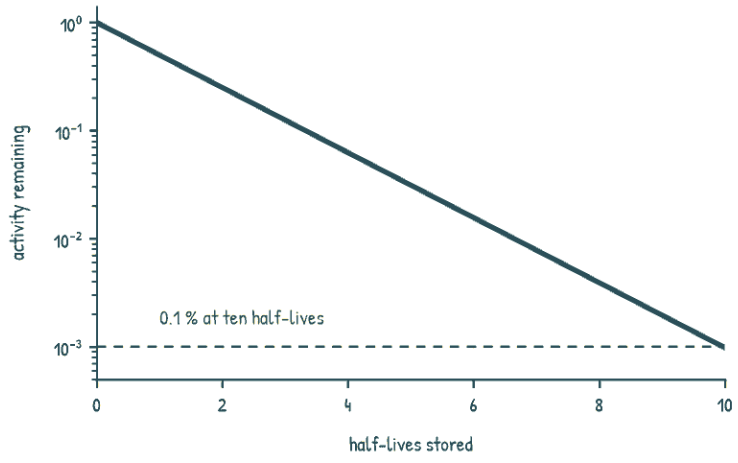
KEY RESULT
the numbers
1.05 kg fissioned per GW·day (thermal).
≈26 kg of 4 % fuel fed per day.
$2^{-10} \approx 0.1\,\%$.

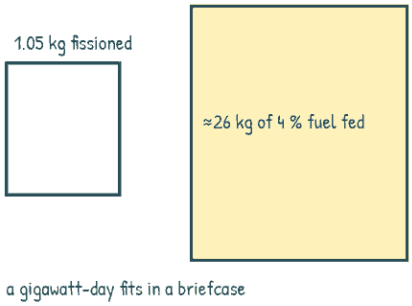
GATE
what this ignores
Burn-up, conversion ratio, decay heat, and the activity of the fission products — second-order for fuel mass, first-order for safety.

TRAP
thermal versus electric
Quoting 1.05 kg/day for a 1 GW *electric* plant. At 33 % efficiency it is three times that.

1.05 kg fissioned per GW·day; the fuel fed is tens of times more because the enrichment is small

the storage rule of thumb




a gigawatt-day fits in a briefcase

KEEP / 1.05 kg per day per GW thermal; ten half-lives leaves 0.1 %

# How long must you count to know the rate to 1 %?

? Radioactive counts arrive like coin flips. The square-root tax sets every measurement time in the lab.

## 1 Poisson statistics in one line

Counts in a fixed window are independent events with a fixed mean, so the variance equals the mean. The absolute uncertainty is √N and the fractional uncertainty is 1/√N — the only error bar that gets relatively smaller as you count more.

$$\sigma_N = \sqrt{N}, \qquad \frac{\sigma_N}{N} = \frac{1}{\sqrt{N}}$$

→ 10 000 counts carry 1 %; 100 counts carry 10 %

## 2 Plan the time backwards from the precision

At 100 counts per minute, 1 % needs 10 000 counts, which is 100 minutes. A weak sample at 1 count per minute needs 10 000 minutes — a week of counting for the same sentence.

$$N = \left(\frac{1}{\epsilon}\right)^2 = 10^4 \text{ for } \epsilon = 1\,\%, \qquad t = \frac{N}{r}$$

→ to halve the error you need four times the counts, hence four times the time

## 3 Dead time and background bend the rule

A Geiger counter is blind for a short dead time after each pulse, so at high rates it undercounts; the correction is n/(1 − nτ_d). Background counts add in quadrature, so a weak source must be counted with and without.

$$n_{\text{true}} = \frac{n_{\text{obs}}}{1 - n_{\text{obs}}\tau_d}, \qquad \sigma_{\text{net}} = \sqrt{N_s + 2N_b}$$

→ at 1 % dead time and $10^4$ counts per second the correction is already 1 %

## 4 A worked counting plan

A source gives 100 counts per minute. Time for 1 %?

$$N = \frac{1}{(0.01)^2} = 10^4 \;\Rightarrow\; t = \frac{10^4}{100} = 100 \text{ min}$$

$$\text{for } 0.5\%: \; N = 4 \times 10^4, \; t = 400 \text{ min}$$

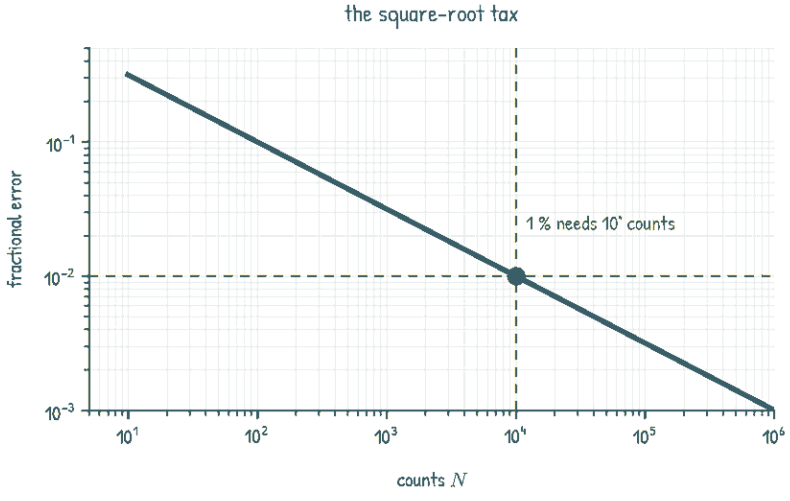
→ four times the time for half the error; the tax is unavoidable

KEY RESULT
the tax
$\sigma/N = 1/\sqrt{N}$
1 % → $10^4$ counts.
4× time halves the error.

GATE
when Poisson holds
Independent decays, a stable rate, and a window short compared with the half-life.

TRAP
the doubled background
Subtracting background without adding its uncertainty — the net error is √(N_s + 2N_b), not √N_s.

σ = √N; to halve the fractional error you need four times the counts and four times the time




rate 100 min⁻¹

1 % → 100 min

0.5 % → 400 min

0.1 % → $10^4$ min

four times the time halves the error

KEEP / σ = √N, so 1 % needs $10^4$ counts and four times the time halves the error

0.1 eV 1 MeV 10 MeV

# How do you date a rock without knowing what it started with?

? Plot one isotope ratio against another. Every mineral of the same rock falls on one line, and the slope is the age.

### 1 Write the daughter as initial plus produced

The daughter isotope present now is whatever was there at birth plus what the parent has since produced. Dividing by a stable reference isotope turns abundances into ratios that can be measured precisely.

$$\frac{D}{S} = \left(\frac{D}{S}\right)_0 + \frac{P}{S}(e^{\lambda t} - 1)$$

→ D/S and P/S are measured; the unknowns are the intercept and the slope

### 2 The line removes the unknown

Different minerals in one rock have different P/S but the same age and the same initial D/S. Plotting D/S against P/S gives a straight line for every mineral: its slope is e^{λt} − 1 and its intercept is the initial ratio.

$$\text{slope} = e^{\lambda t} - 1 \;\Rightarrow\; t = \frac{\ln(1 + \text{slope})}{\lambda}$$

→ scatter off the line is the diagnostic: it says the rock was disturbed

### 3 Read an age off a slope

For Rb-Sr, λ = 1.42×10⁻¹¹ per year. A slope of 0.05 gives ln(1.05)/λ = 3.4×10⁹ years, which is the age of an Archean rock.

$$t = \frac{\ln 1.05}{1.42 \times 10^{-11}} = 3.4 \times 10^{9}\ \text{y}$$

→ the same arithmetic as carbon dating, with a different constant and no initial-ratio guess

### 4 Two minerals, one slope

Rb-Sr, λ = 1.42×10⁻¹¹ y⁻¹. Mineral A: 2.0 and 0.720. Mineral B: 5.0 and 0.750.

$$\text{slope} = \frac{0.750 - 0.720}{5.0 - 2.0} = 0.0100$$

$$t = \frac{\ln 1.010}{1.42 \times 10^{-11}} = 7.0 \times 10^{8}\ \text{y}$$

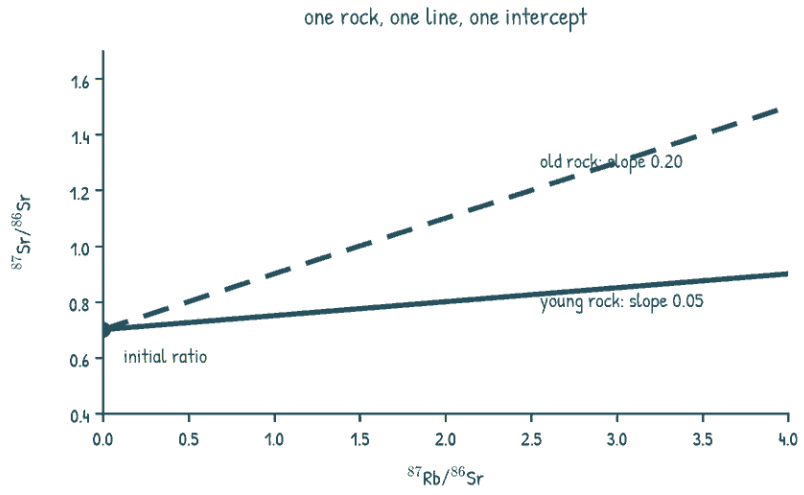
→ check the intercept: 0.720 − 0.0100(2.0) = 0.700, the same for B — one age, one initial ratio

KEY RESULT
the line
D/S against P/S.
Slope = e^{λt} − 1.
Intercept = initial ratio.

GATE
the isochron assumptions
One age for all minerals, one initial ratio, a closed system since formation.

TRAP
the model age
Using a single mineral and an assumed initial ratio gives a model age, not an isochron age. It is only as good as the assumption.

Plot daughter/reference against parent/reference: slope gives the age, intercept the start




$\text{slope} = e^{\lambda t} - 1$

slope 0.05 → 3.4 Gyr

intercept = initial ratio

scatter off the line = disturbed rock

KEEP / slope = e^{λt} − 1; the intercept absorbs the unknown initial daughter

0.1 eV — 1 MeV ▼ — 10 MeV

# Why do slow neutrons diffract from crystals?

? The same wave logic as the electron: compare the wavelength with the spacing. The spacing that matters is the lattice's.

### 1 A thermal neutron has an angstrom wavelength

A 25 meV neutron has λ = h/√(2mE) = 1.8 Å — the scale of atomic planes. That is why neutron diffraction needs no accelerator.

$$\lambda = \frac{h}{\sqrt{2m_n E}} = 1.8\ \text{Å} \quad \text{at } 25\ \text{meV}$$

→ the same formula as the electron's 1.226/√V nm, with the neutron mass and energy

### 2 Two rulers, one formula

Inside a nucleus the nucleon spacing is about 1.8 fm, a thousand times smaller. A thermal neutron with a 1.8 Å wave cannot resolve that structure — it sees the whole nucleus as a point scatterer. The physics did not change; the ratio λ/d did.

$$d_{\text{nucleon}} \approx \left(\frac{1}{0.17\ \text{fm}^{-3}}\right)^{1/3} \approx 1.8\ \text{fm} \ll \lambda$$

→ high-energy electrons (pm waves) are needed to see inside a nucleus, for exactly this reason

### 3 Why neutrons rather than X-rays

X-rays scatter from electrons, so hydrogen is nearly invisible. Neutrons scatter from nuclei, so light atoms show and isotopes contrast — same Bragg law, a different probe.

$$2d\sin\theta = n\lambda \quad (\text{Bragg}), \qquad \lambda = 1.8\ \text{Å} \Rightarrow \theta \text{ of order } 10^\circ$$

→ magnetic moments also scatter neutrons, which is why they map spin structures

### 4 A worked wavelength

A 25 meV neutron, and the energy needed to reach 1 fm.

$$\lambda = \frac{h}{\sqrt{2m_n E}} = 1.81\ \text{Å} \quad \text{at } 25\ \text{meV}$$

$$E_k = \sqrt{(pc)^2 + (m_n c^2)^2} - m_n c^2 = 616\ \text{MeV} \quad \text{for } \lambda = 1\ \text{fm}$$

→ at that momentum the non-relativistic $h^2/2m\lambda^2$ would say 818 MeV — the relativistic form is the one to use

KEY RESULT
the comparison
λ(25 meV) = 1.8 Å.
Lattice spacing ≈ 1.8 Å → diffraction.
Nucleon spacing 1.8 fm → point scatterer.

GATE
when diffraction works
λ comparable to the spacing. Much shorter gives a point image; much longer gives no Bragg condition at all.

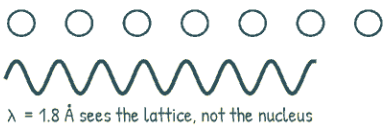
TRAP
mixing the rulers
Using a thermal neutron to probe nuclear structure, or an MeV neutron to probe a crystal — the wavelength is wrong in both directions.

Match the wavelength to the spacing: thermal neutrons see crystals, high-energy probes see nuclei

nucleus: spacing 1.8 fm



crystal: spacing 1.8 Å



λ = 1.8 Å sees the lattice, not the nucleus

25 meV → 1.8 Å

616 MeV → 1 fm

non-relativistic: 818 MeV (wrong)

$2.5 \times 10^{10}$ in energy

same λ = h/p, different probe

KEEP / 1.8 Å neutrons match atomic spacings; inside a nucleus the same wave sees a point

0.1 eV — 1 MeV ▼ — 10 MeV

# How long can the Sun keep shining?

? Ten per cent of its hydrogen, 0.7 per cent of the mass of that fuel, and a luminosity you can measure.

### 1 Count the fuel, then the energy per gram

Only the core hydrogen burns, and only 0.7 % of the fused mass becomes energy. Multiplying the available mass by 0.007c² gives the total energy budget; dividing by the luminosity gives the lifetime.

$$E \approx 0.1(0.7M_\odot)(0.007c^2) = 8.7 \times 10^{43}\ \text{J}$$

→ the numbers: 10 % of the hydrogen, 70 % hydrogen by mass, 0.7 % mass-to-energy

### 2 Divide by the luminosity

The Sun radiates 3.83×10²⁶ W, so the budget lasts about 7×10⁹ years on the main sequence — and the independent mass-loss rate L/c² times that lifetime returns the burned mass, a useful cross-check.

$$t = \frac{8.7 \times 10^{43}}{3.83 \times 10^{26}} = 2.3 \times 10^{17}\ \text{s} \approx 7 \times 10^{9}\ \text{y}$$

→ the Sun is middle-aged; the first weak step in the p-p chain, not the fuel, sets the pace

### 3 The same liquid drop says where the table ends

The fission barrier vanishes when the Coulomb gain from deformation beats twice the surface cost, near Z²/A ≈ 50. With A ≈ 2.5Z that is Z ≈ 125; shell corrections prop up superheavies a little beyond, but not indefinitely.

$$\frac{Z^2}{A} \approx 50, \quad A \approx 2.5Z \ \Rightarrow\ Z \approx 125$$

→ bismuth (83) is the last fully stable element; beyond it, alpha decay is the leak

### 4 A worked solar budget

Ten per cent of the Sun's hydrogen, 0.7 % mass-to-energy.

$$E = 0.1(0.7)(1.989 \times 10^{30})(0.007)(9 \times 10^{16}) = 8.8 \times 10^{43}\ \text{J}$$

$$t = \frac{8.8 \times 10^{43}}{3.83 \times 10^{26}} = 2.3 \times 10^{17}\ \text{s} = 7.3 \times 10^{9}\ \text{y}$$

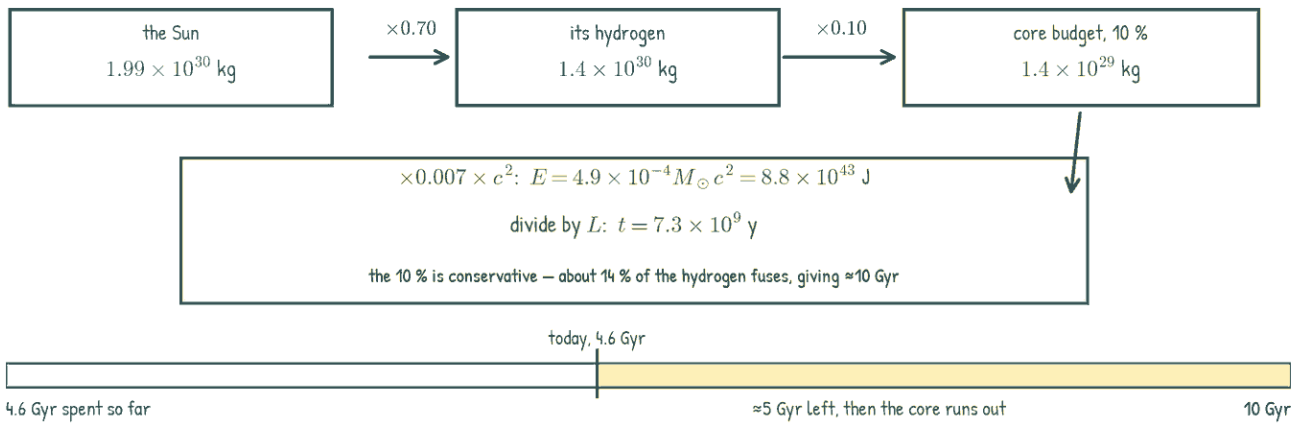
→ the Sun has burned about half its hydrogen; it is middle-aged, not ancient

KEY RESULT
the budget
10 % of the hydrogen is available.
0.007 of the mass becomes energy.
L = 3.83×10²⁶ W → ≈7×10⁹ y.

GATE
what is left out
Mass loss, core mixing, the red-giant phase, and the fact that the last of the hydrogen burns inefficiently.

TRAP
all the hydrogen
Using the whole mass, or the whole hydrogen, in the budget overestimates the lifetime by a factor of ten.

0.1 × 0.7 × 0.007 × M☉c² divided by L gives about 7 billion years




KEEP / 0.07 × 0.1 M☉ × 0.007c² / L ≈ 10 Gyr; and the drop says where the table ends